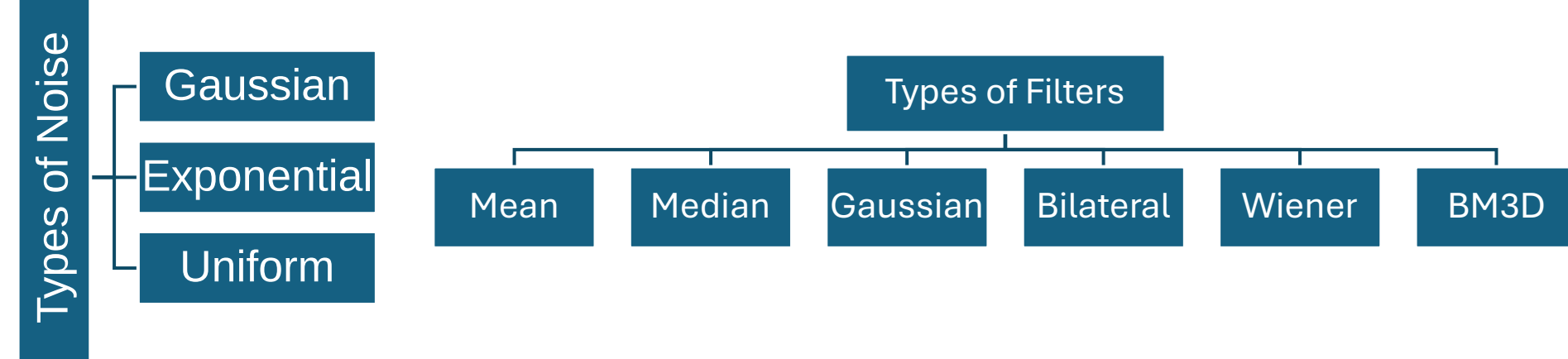


Overview and Goals

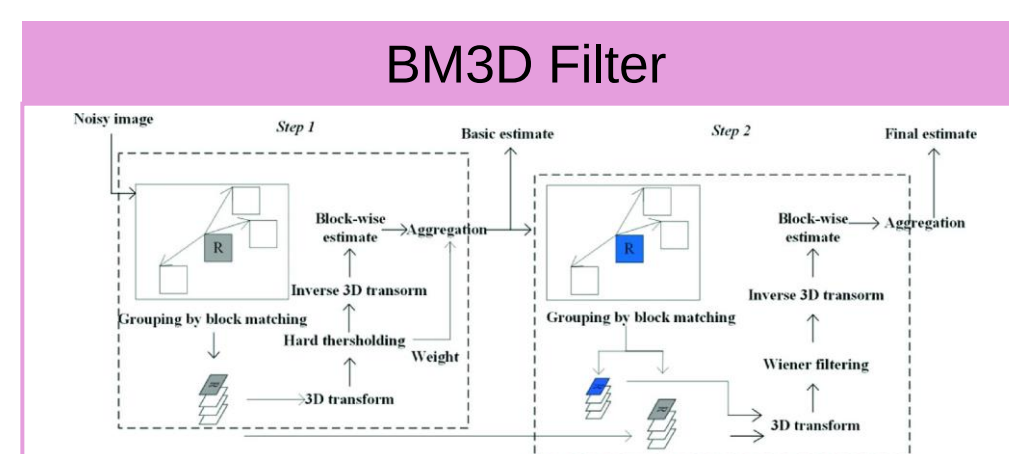
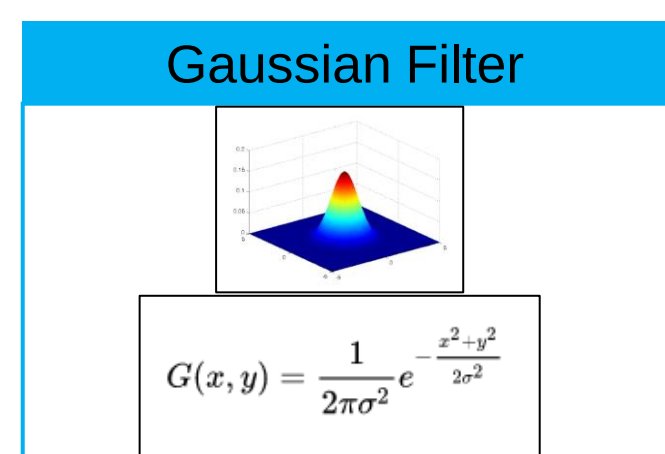
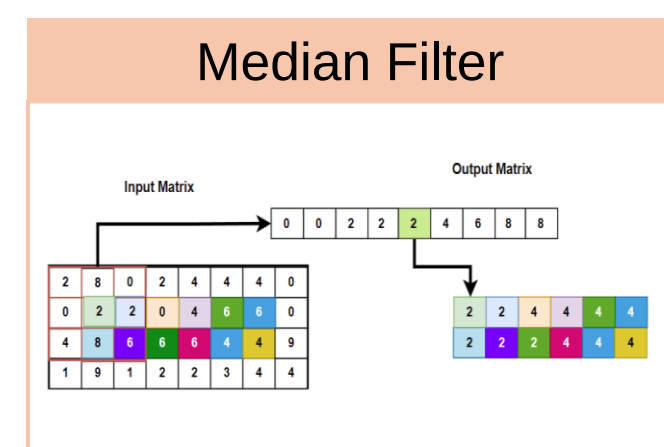
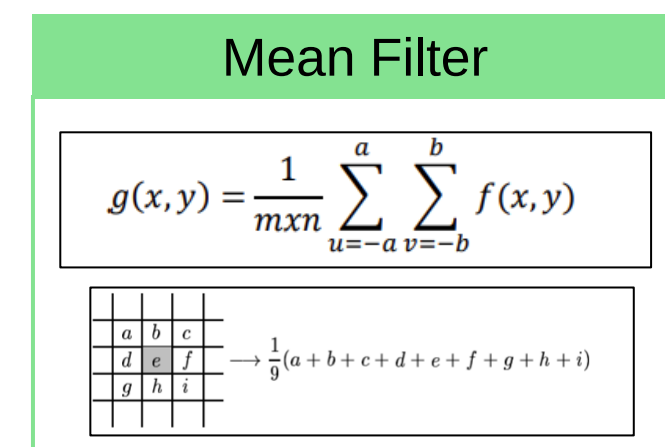
- The denoising of microscopic images is essential for maintaining the authenticity and preserving the originality of the data. Whether it is for the diagnosis of medical conditions or the study of microorganisms, effective denoising enhances performance.
- Our objective is to apply a non-deep learning method to denoise a dataset, establishing a baseline performance. Subsequently, employ a deep learning method to denoise the same dataset, aiming to surpass the established baseline.

Non DL-Method

- Initially, determine the type of noise to be added to the image. Noise is introduced in the form of certain statistical distributions that mimic real noise encountered during the capture of microscopic images.
- The manifestation of these noises, including their intensity, depends on various factors such as the skill of the operator, capturing conditions, and the quality of the microscope. However, we assume a certain intensity level for consistency.



- Subsequent to the addition of noise, the images are denoised using traditional image processing techniques. The selection of approaches or filters is primarily influenced by the specific application and the dataset at hand.
- Given the common usage and availability of these noises and filtering techniques, we have structured the process to allow each noise type to be individually added to a single image. Each noisy image is then denoised using all the aforementioned filters.
- Performance is evaluated by computing the and Peak Signal-to-Noise Ratio (PSNR) for all noisy and denoised images, and the results are documented in an Excel sheet for easy analysis.



Selection of Deep Learning Methods

Background and Motivation:

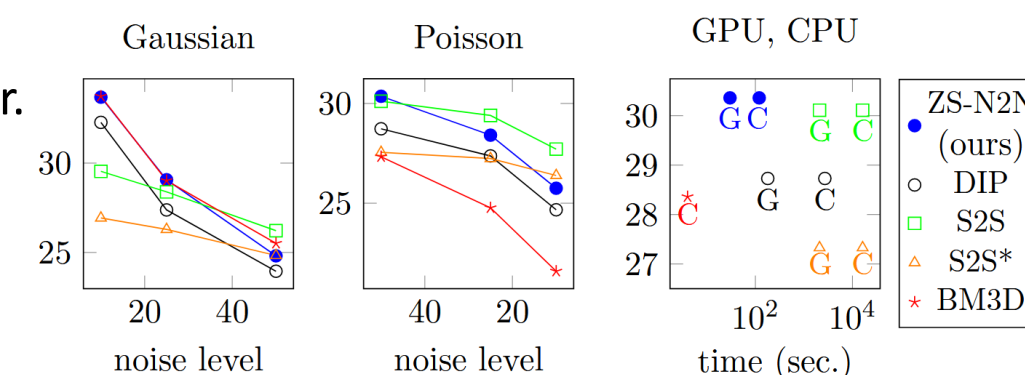
- Traditional image denoising methods require large datasets of clean images, which can be expensive and time-consuming to obtain.
- Zero-Shot Noise2Noise (ZS-N2N) is selected due to its ability to perform high-quality image denoising without any training data, noise model, or noise level information.

Approach:

- Inspired by Noise2Noise and Neighbor2Neighbor.
- Utilizes a simple 2-layer neural network.
- Applies consistency loss for improved performance.

Comparison with Other Methods:

- Traditional methods like BM3D require noise level information and fail to generalize to different noise types.
- Other deep learning methods (e.g. Self2Self) require either specific noise models or are computationally intensive.



Workflow of the DL Method

Load an Image:

- Load and resize clean image to a uniform size.

Adding Noise:

- Select either Gaussian or Poisson or Uniform noise to the clean image to create noisy images.

Downsampling :

- Convolve the noisy image with two fixed filters to clean two downsampled images

Pairing:

- Train the network to map one downsampled image to the other using a consistency loss

Training:

- Use a simple 2-layer network with only 20k parameters

Prediction:

- The network predicts the noise for each image pair.

Prevent Overfitting:

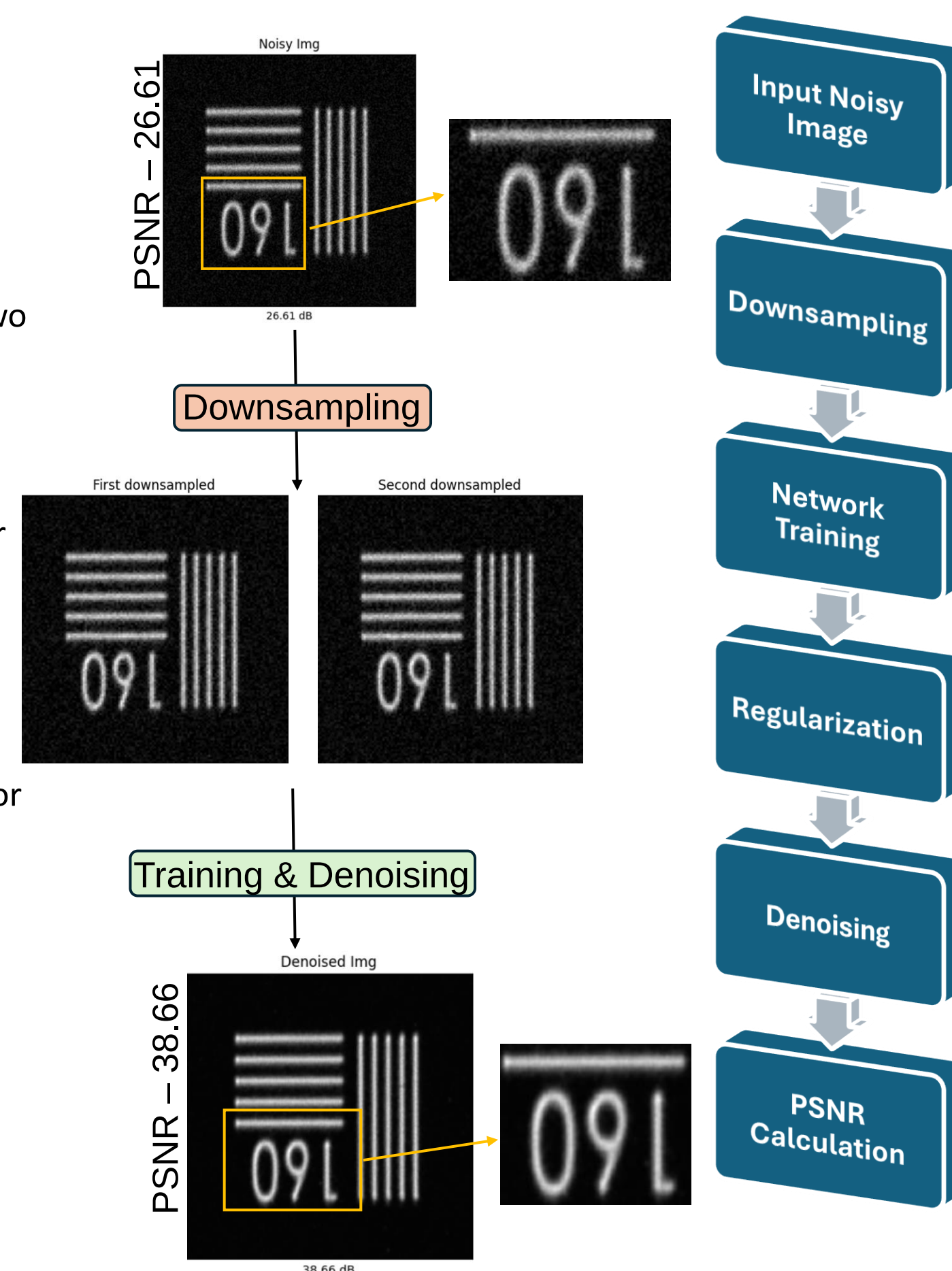
- Apply regularization techniques prevent overfitting.

PSNR :

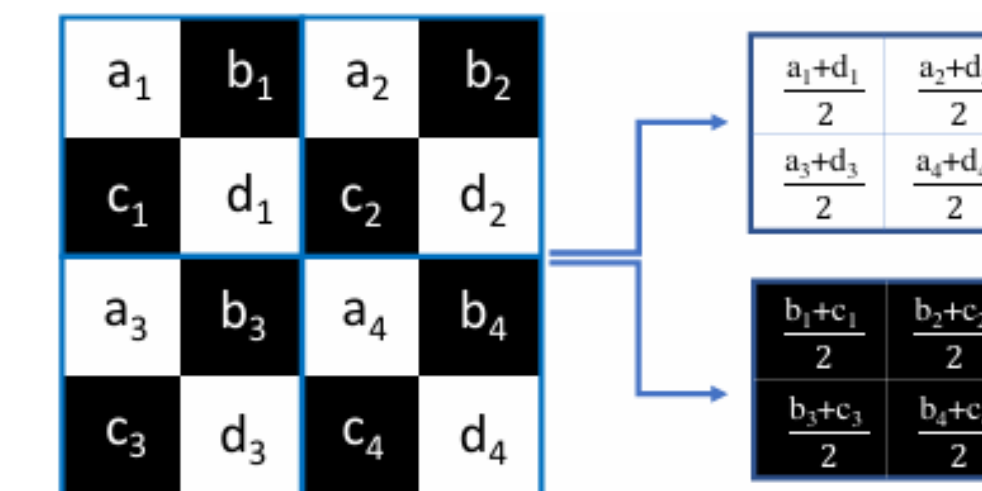
- Calculate Peak Signal-to-Noise Ratio (PSNR) to evaluate image quality of the denoised image.

Visualization:

- Compare clean, noisy, and denoised images.



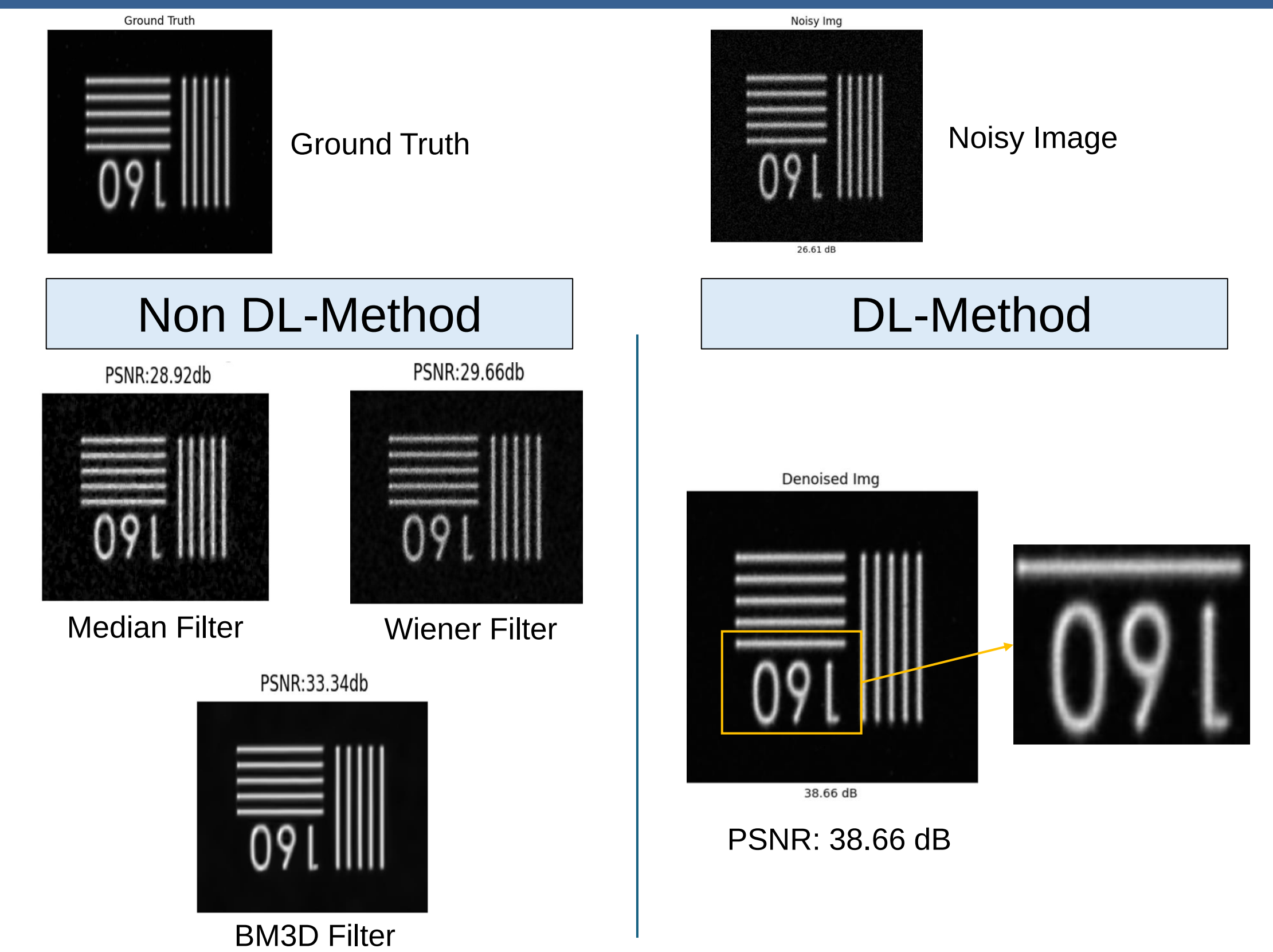
Downsampling



- The Image Pair Downsampler decomposes an image into two images of half the spatial resolution by averaging diagonal and anti-diagonal pixels of 2x2 non-overlapping patches to create two low-resolution images.

- In the above example the input is a 4x4 image, and the output is two 2x2 images.

Results and Analysis



Conclusions

- This method uses a 2-layer network, downsampling noisy images into pairs via convolution. The network is trained to map one down sampled image to the other, achieving high-quality denoising with low computational cost, generalizing across various noise types.

Results	Noisy Image	Non-DL Method			DL Method
		Median	Wiener	BM3D	
PSNR (dB)	26.61	28.92	29.66	33.34	38.66